Macroeconomic Modeling at the Federal Reserve Board

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Federal Reserve Board

NBER

Note: Views are my own and do not necessarily reflect those of the Federal Reserve Board or the Federal Reserve System.

Thanks to my colleagues for inputs.

Nonconfidential // External

Macro modeling at the Federal Reserve Board

- Produce internally consistent projections
  - for activity, inflation, labor markets, and interest (policy) rates
- Quantify counterfactuals: "what if policy/shocks/assumptions differ?"
- Support risk management: scenarios, uncertainty bands, event probabilities
- Support evaluation of policy strategies
- Highlight mechanisms and help policy deliberation / communication

Portfolio of models used at the Board

- Large-scale macroeconometric:
  - FRB/US (and derivatives) for baseline/scenarios, risk analysis, and policy strategy sims
- DSGE suite, for structural interpretation, policy experiments, other focused questions:
  - Medium-scale DSGEs (including a HANK model)
  - Specialized DSGEs (e.g., financial frictions, labor market frictions)
- International / multi-country structural models:
  - Multi-country DSGEs (e.g., SIGMA / GEMUS) for spillovers and global scenarios
- Auxiliary empirical / semi-structural models:
  - Reduced-form time-series tools (VARs, factor models) for forecasting / decompositions
  - State-space / filtering systems for trends and latent objects
    - Potential, NAIRU, trend or 'underlying' inflation, etc

The regular policy workflow (Tealbook A)

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- For each FOMC meeting, staff prepare:
  - A baseline projection (judgmental conditioning assumptions + model inputs),
  - Counterfactual scenarios reflecting salient risks (alternative shocks/assumptions),
  - Uncertainty assessments (bands/probabilities), often via linear tools and stochastic simulations,
  - Assessment of monetary policy strategies

Tealbook A example (Jan 2020): FOMC20200129tealbooka20200117.pdf

The FRB/US universe: core components

- FRB/US: large-scale (∼290 eqns, ∼370 variables) nonlinear quarterly model of U.S.
  - Behavioral equations estimated largely equation-by-equation ▸ example
  - National-accounts-consistent structure with explicit policy/financial channels
- VAR expectations system: forecasting structure used to build expectations objects
  - Used in estimation and simulation
  - Not the only expectations option (can be model consistent or a mix) ▸ details
- Supply-side state-space model: auxiliary filter used to estimate trending components
  - Potential, NAIRU, trend participation/workweek, etc. ▸ details
- Public version: code, documentation, programs, and database
  - Including a forecast consistent with policymakers' SEP

See: federalreserve.gov/econres/us-models-about.htm

The FRB/US universe: linear tools for other applications

- Solving the nonlinear model with shooting algorithms can be time-consuming
  - Especially with model-consistent expectations
- LINVER: linearized FRB/US, efficiently solved with standard solvers (e.g. Dynare)
  - Example later of analyzing effective lower bound on nominal interest rates
- sFRB: smaller linearized model designed for complex monetary-policy experiments
  - Parameters chosen to match key FRB/US impulse responses

LINVER public page: federalreserve.gov/econres/us-models-linver.htm

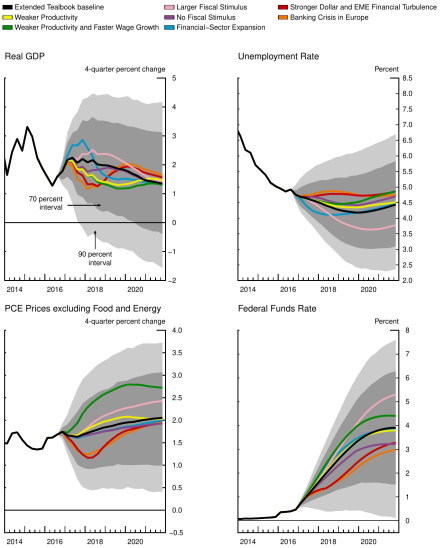
Example use: FRB/US counterfactuals for fiscal policy (Dec 2016)

- Example: fiscal policy tax cuts expected over 2017
- Baseline projection
  - Estimate of the fiscal impetus, using elasticities from literature
- A set of policy/scenario alternative counterfactuals using FRB/US
  - Larger fiscal stimulus: larger package / different composition than baseline
  - No fiscal stimulus: baseline assumed tax cut does not materialize
- Fiscal scenarios shift the paths for GDP growth, unemployment, inflation, and the funds rate relative to the baseline

Source: Tealbook A (Dec 2016), Risks & Uncertainty, *Forecast Confidence Intervals and Alternative Scenarios*.

Forecast Confidence Intervals and Alternative Scenarios

Confidence Intervals Based on FRB/US Stochastic Simulations



Risks & Uncertainty

Another example: "Stronger Aggregate Supply" as an upside risk

- Risks & Uncertainty section: Even with solid demand, the economy may be able to grow faster *without* generating inflation pressure if potential output is higher than assumed.
  - Higher effective labor supply / participation:
    - Trend LFPR declines more slowly (so LFPR rises rel. to baseline), increasing potential hours
  - Lower structural slack / NAIRU:
    - A lower natural rate of u/e implies more sustainable employment at a given inflation rate
  - Faster structural productivity growth:
    - TFP (underlying productivity) growth assumed higher than baseline
- Implication: faster output growth, higher real incomes, and less inflation pressure (for a given demand path) because resource utilization is lower than in the baseline

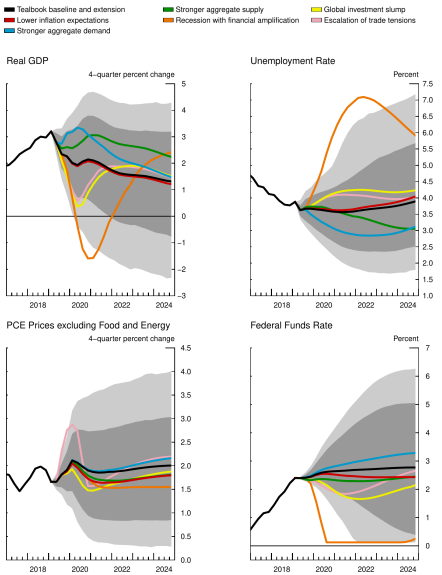
Public Tealbook example:

- Jul 2019 Tealbook A, Risks & Uncertainty (FRB/US-based): "Stronger Aggregate Supply" scenario

Risks & Uncertainty

Forecast Confidence Intervals and Alternative Scenarios

Confidence Intervals Based on FRB/US Stochastic Simulations*



Long-run assumptions about r^{LR}

- The longer-run equilibrium real funds rate: the real rate consistent with the economy at full potential once cyclical shocks abate

- A staff assumption / anchor used in the suite of models and as the intercept in policy rules, e.g.

$$R_t \;=\; r^{LR} + \pi_t \;+\; 0.5\left(\pi_t - \pi^{LR}\right) \;+\; \text{output gap}_t$$

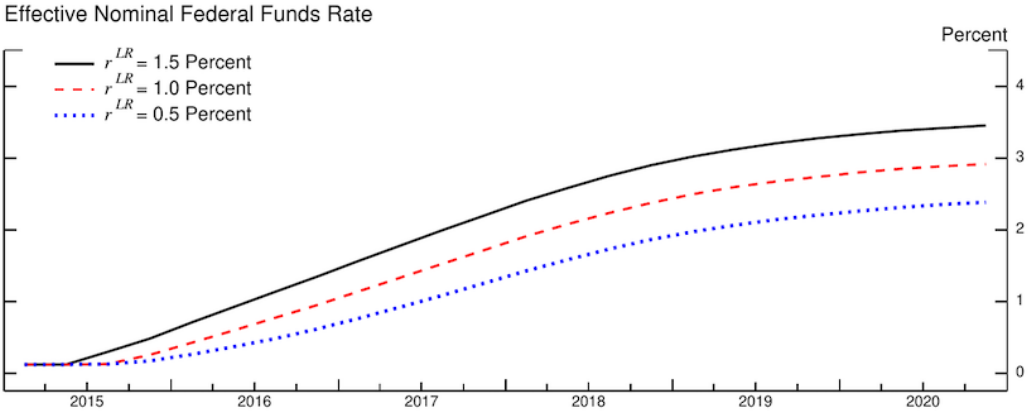
- Updated over time: Tealbook B (Jan 2016) notes staff had, in several steps since June 2014, *lowered* its longer-run equilibrium real rate assumption from 2% to $1\frac{1}{4}$%.

- Interpretation: consistent with econometric evidence of a decline in low-frequency determinants (e.g., trend growth) ▸ econometric evidence
  - also consistent with declining longer-run SEP implied real-rate measures

Source: *Tealbook B, Jan 21 2016*, box "The Equilibrium Real Rate in the Longer Run."

Alternative assumptions about r^{LR}

From the June 2015 Tealbook B:



Risks & Uncertainty

Forecast Confidence Intervals and Alternative Scenarios

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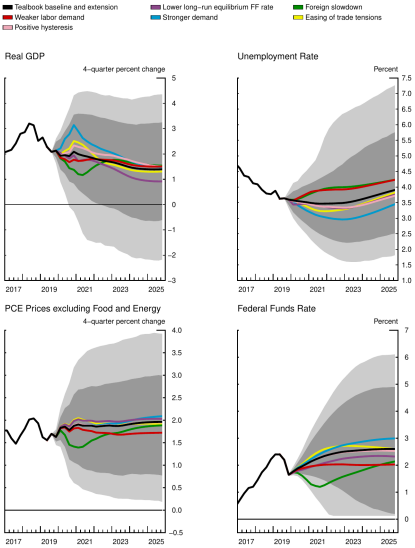


* The dark gray shaded area is the 70 percent interval, and the light gray shaded area is the 90 percent

Stochastic sims: capturing recessions and "effective lower bound" risks

- Motivation: business cycles are asymmetric (deep recessions vs mild expansions)
- Naive bootstraps under-generate severe downturns
- Approach:
  - classify periods into regimes (non-recession, conventional recession, Great Recession),
  - assume Markov transitions calibrated to match frequencies of regimes,
  - when entering recession, replay a randomly selected historical recession shock sequence.
- Useful when analyzing tail risks and "lower bound on interest rates" episodes
- The framework review memo on Monetary Policy Strategies (Chung, Jones, Lepetit, Martin, 2025) used this heavily

Software and public access

- Internal workflow spans multiple platforms:
  - EViews (mature nonlinear MCE solvers; user-friendly for time series),
  - Python (addresses extensibility, ecosystem, and programming constraints),
  - Matlab (used for LINVER and sFRB workflows, including ELB implementation for linear RE models)
- Public release (since 2014) enables outside use

Conclusions

- Macroeconomic modeling extensive at the Board
- FRB/US is a policy workhorse: broad coverage + realistic transmission channels + repeated counterfactual capability.
- The "universe" matters: expectations systems, supply-side filters, and linearized versions are essential complements.
- Tealbooks (with some delay) provide a public window into how projections, scenarios, and uncertainty are organized for policy deliberation.

Extra

Supply-side structure: production + trends via state-space filtering

- Business-sector production typically modeled with Cobb–Douglas technology
- A satellite state-space model estimates trend components:
  - potential output, NAIRU, trend participation, trend workweek, trend productivity
- These latent objects support "slack" measures and medium-run projections

Example: consumption

Long-run (equilibrium/target) relationship:

$$c_t^* = \alpha_0 + \alpha_1 y_t + \alpha_2 w_t + \alpha_3 r_t + \alpha_4 z_t$$

- c_t^*: log "target" consumption level
- y_t: log disposable income (or labor income)
- w_t: log wealth (e.g., household net worth)
- r_t: real interest rate (or user cost / financial conditions index)
- z_t: other controls (taxes/transfers, credit conditions, sentiment, etc.)

Short-run dynamics (estimated adjustment):

$$\Delta c_t = \beta_0 + \beta_1 (c_{t-1} - c_{t-1}^*) + \sum_{i=1}^{p} \beta_{2i} \Delta c_{t-i} + \sum_{j=0}^{q} \beta_{3j} \Delta y_{t-j} + \varepsilon_t$$

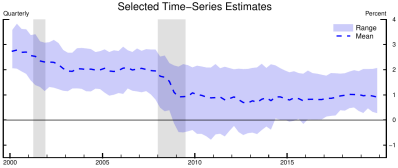
- $\beta_1 < 0$ is the speed of adjustment back toward the long-run target
- Lags allow inertia, income terms capture near-term propagation
- This equation can be augmented with expected future terms (e.g., expected income)

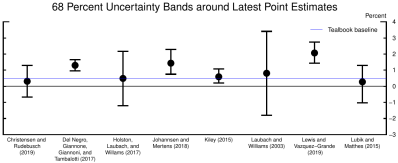
Expectations in the model

- Expectations enter explicitly in many equations
  - Enabling anticipated shocks and forward-looking policy analysis
- Mechanism: polynomial adjustment costs (PAC)
  - Agents gradually adjust toward an "equilibrium goal" implied by expected future conditions
- In simulation, options include:
  - Constrained VAR expectations,
  - Model-consistent expectations (MCE),
  - "Mixed" expectations (some sectors VAR, some MCE)

Monetary Policy Strategies

Estimates of the Equilibrium Real Federal Funds Rate in the Longer Run





Longer–Run Values from Selected Forecasters

	Release Date	Percent
Tealbook baseline	Mar. 2020	.50
Median SEP	Dec. 2019	.50
Median Survey of Primary Dealers	Jan. 2020	.40
Blue Chip consensus	Oct. 2019	.29
Congressional Budget Office	Jan. 2020	.65

Note: The latest time–series estimates in the top and middle panels are from 2019:Q4. The shaded vertical areas in the top panel are NBER recessions. See the appendix for the construction of the values reported in the bottom panel.